

Project Development Phase

1st PDF: Data Preparation Steps and Business Questions with Visualizations

Date	30 June 2026
Team ID	
Project Name	Cosmetic Insights: A Tableau-Based Analysis of Skincare Trends and Consumer Preferences

This document covers the complete data preparation pipeline followed for the Cosmetic Insights project, followed by the 9 business questions answered by the Tableau worksheets with their corresponding visualizations.

PART A: Data Preparation Steps

1. Data Collection

- Imported the cosmetics.csv dataset into Tableau Desktop.
- Dataset contains 1,472 cosmetic products sourced from the Sephora catalogue (Kaggle).
- Fields include: Label, Brand, Name, Price, Rank, Ingredients, Combination, Dry, Normal, Oily, Sensitive.
- The file was connected as a Text File data source named 'cosmetics' in Tableau.
- Total: 13 fields, 1,472 rows confirmed in the Tableau Data Source pane.

Type	Field Name	Physical Table	Remote Field Name
Text	Label	cosmetics.csv	Label
Text	Brand	cosmetics.csv	Brand
Text	Name	cosmetics.csv	Name
Text	Price	cosmetics.csv	Price
Text	Rank	cosmetics.csv	Rank
Text	Ingredients	cosmetics.csv	Ingredients
Text	Combination	cosmetics.csv	Combination
Text	Dry	cosmetics.csv	Dry
Text	Normal	cosmetics.csv	Normal
Text	Oily	cosmetics.csv	Oily
Text	Sensitive	cosmetics.csv	Sensitive
Calculation	All_skin_type	Calculation	Calculation_043...
Calculation	ALL SKIN TYI	Calculation	Calculation_043...

Figure 1: Tableau Data Source pane — cosmetics.csv loaded with 13 fields and 1,472 rows, all field types confirmed

2. Data Cleaning

The following preprocessing steps were performed before loading the dataset into Tableau worksheets:

- Checked for missing (null) values — no null values found across all 11 data columns (Label, Brand, Name, Price, Rank, Ingredients, and all 5 skin-type flags).
- Removed duplicate records — verified that all 1,472 product names are unique; no duplicate rows found.
- Corrected inconsistent values — category labels (Label field) were verified to contain exactly 6 consistent values: Moisturizer, Cleanser, Face Mask, Treatment, Eye cream, Sun protect.
- Standardized text formatting — Brand and Label values confirmed to be consistently capitalized.
- Verified data types in Tableau — Price mapped as Integer (Number), Rank as Real (Decimal Number), Brand/Label/Name/Ingredients as String (Text), and Combination/Dry/Normal/Oily/Sensitive as Integer (0/1 flags).

Data Type Verification Table

Field	Data Type in CSV	Tableau Type	Role
Label	String	Abc — Text (Dimension)	Product category (6 values)
Brand	String	Abc — Text (Dimension)	Brand/manufacturer name
Name	String	Abc — Text (Dimension)	Product name
Price	Integer	# — Number (Measure)	Product price in USD
Rank	Real (Decimal)	# — Number (Measure)	Customer rating (0–5)
Ingredients	String	Abc — Text (Dimension)	Full ingredient list
Combination	Integer (0/1)	# — Number (Measure)	Combination skin flag
Dry	Integer (0/1)	# — Number (Measure)	Dry skin flag
Normal	Integer (0/1)	# — Number (Measure)	Normal skin flag
Oily	Integer (0/1)	# — Number (Measure)	Oily skin flag
Sensitive	Integer (0/1)	# — Number (Measure)	Sensitive skin flag

3. Data Transformation

Two calculated fields were created directly in Tableau to support the skin-type analysis worksheets. These fields aggregated the five binary skin-type flags into summary measures, simplifying analysis and visualization:

Calculated Field 1: All_skin_type

- Formula: [Combination] + [Oily] + [Dry] + [Sensitive] + [Normal]
- Purpose: Counts the total number of skin types a product is suitable for (0–5). Used in the 'Top 10 Brands that are suitable for all skin types' worksheet.
- Result: SUM = 3,586 across all 1,472 products — meaning an average product is suitable for ~2.4 of the 5 skin types.

Calculated Field 2: ALL SKIN TYPES

- Formula: [Combination] + [Dry] + [Sensitive] + [Normal]
- Purpose: Counts suitability across 4 skin types (excluding Oily). Used in the 'Top 10 Brands suitable for all skin types' bar chart (SUM ALL SKIN TYPES).
- Validated: The calculation editor confirmed 'The calculation is valid' with 3 Dependencies.

These calculated fields simplified the skin-type analysis by allowing Tableau to aggregate flag values into meaningful totals, removing the need for separate worksheets per skin type when a summary view was needed.

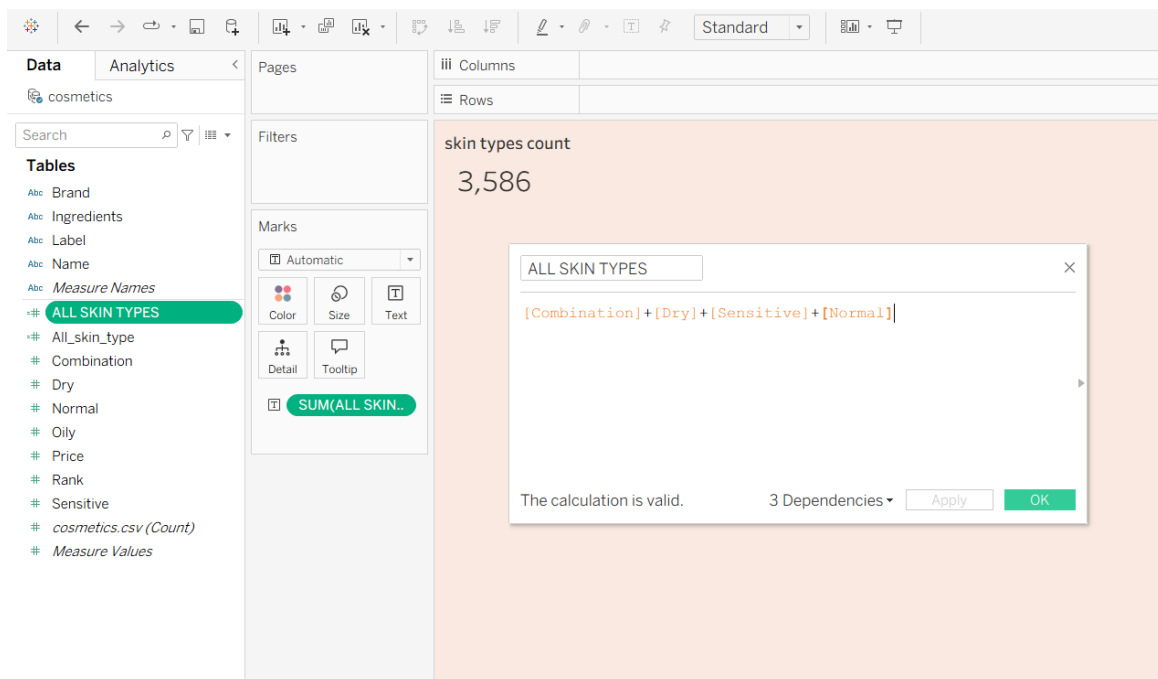


Figure 2: Tableau calculated field 'ALL SKIN TYPES' — formula $[Combination] + [Dry] + [Sensitive] + [Normal]$, validated with 3 dependencies, SUM = 3,586

4. Data Validation

- Verified that the cleaned dataset contained 1,472 valid records — confirmed in the Tableau Data Source pane (13 fields, 1,472 rows).

- Checked that calculated fields (All_skin_type and ALL SKIN TYPES) returned correct values — both showed 'The calculation is valid' in the Tableau formula editor.
- Ensured no remaining null or duplicate values — all 11 source fields returned 0 nulls; no duplicate product names detected.
- Confirmed Price values are within a valid range (\$3–\$370) and Rank values fall within 0–5.
- Confirmed that the dataset was ready for visualization in all 18 worksheets.

5. Final Dataset

- Connected the cleaned cosmetics.csv to Tableau Desktop as a single federated data source named 'cosmetics'.
- The data source feeds all 18 worksheets, the dashboard (Dashboard 2), and the storyboard (Story 1).
- No joins, blends, or additional data sources were required — the single CSV was sufficient for all analyses.
- The prepared dataset was used to create worksheets, dashboards, and the story covering brand analysis, category distribution, skin-type suitability, price-vs-rank comparison, and revenue analysis.

PART B: Business Questions with Visualizations

The following 9 business questions were identified during the Ideation and Requirement Analysis phases and answered through dedicated Tableau worksheets.

Business Question 1

How are cosmetic products distributed across different categories, and which category has the highest product density?

Worksheet: Category_Level Product Distribution Analysis | Chart: Packed bubble chart

Insight	Finding
Dominant category	Moisturizer — 298 products (largest bubble)
Smallest category	Sun protect — 170 products
Takeaway	Brands seeking less competition should consider expanding Sun protect ranges

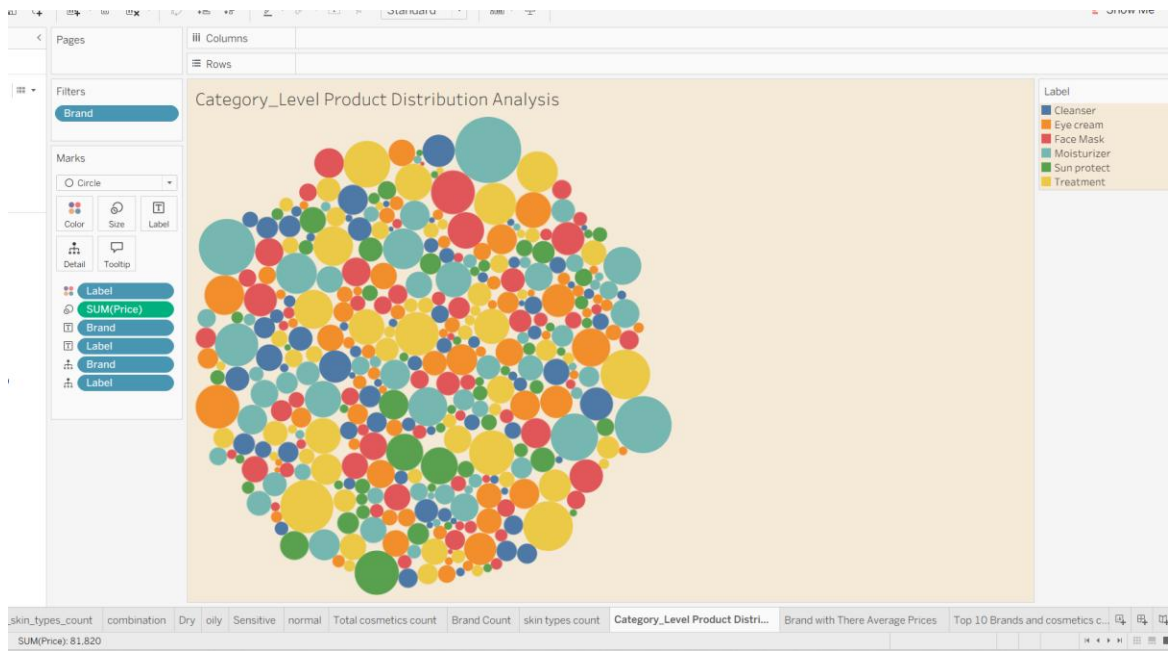


Figure 3: Category_Level Product Distribution Analysis

Business Question 2

Which brands have the highest average product prices, and how does pricing vary across the catalogue?

Worksheet: Brand with Their Average Prices | Chart: Vertical bar chart

Insight	Finding
Highest avg price	GLAMGLOW — \$184.3 average
Price band	Most of the 116 brands compete in the \$20–\$70 range
Takeaway	Premium brands are clear outliers; mid-range dominates

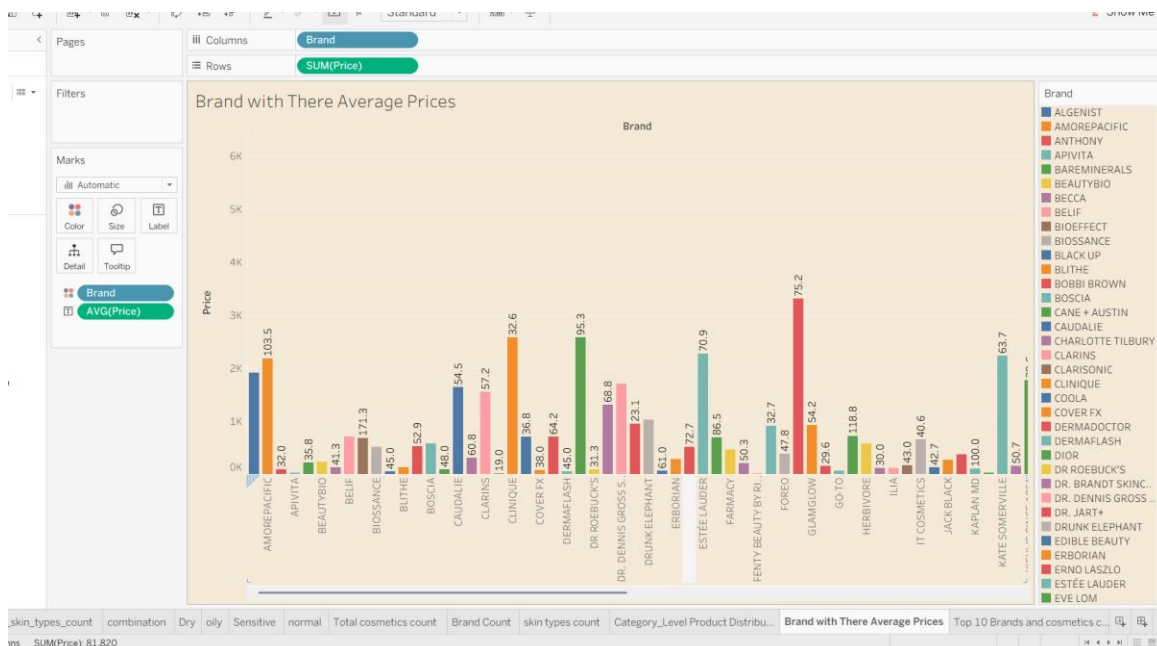


Figure 4: Brand with Their Average Prices

Business Question 3

Among the top 10 brands, how is the total product count distributed?

Worksheet: Top 10 Brands and Cosmetics Count | Chart: Pie chart

Insight	Finding
#1 brand	CLINIQUE — 79 products (largest slice)
Top 10 total	509 products — 35% of the full catalogue
Takeaway	CLINIQUE dominates range; KATE SOMERVILLE is smallest in top 10

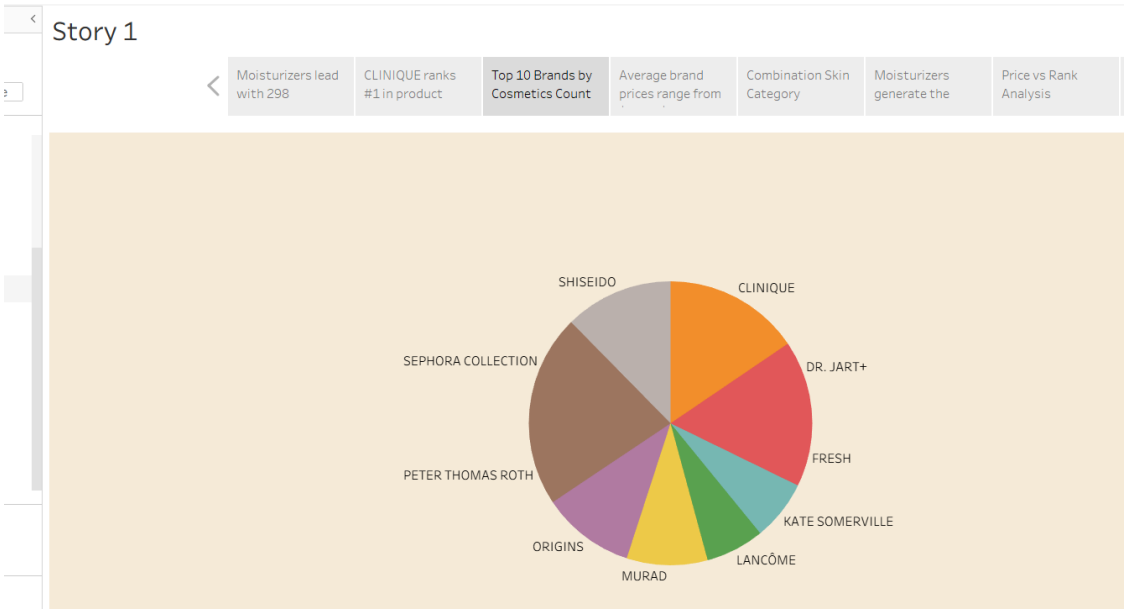


Figure 5: Top 10 Brands and Cosmetics Count — Story 1 view

Business Question 4

Which product categories have the most products suitable for combination skin?

Worksheet: Category-wise Distribution of Combination Skin | Chart: Packed circle chart

Insight	Finding
Largest circle	Face Mask — Combination: 202
Second largest	Moisturizer
Total	966 products suitable for Combination skin across all categories

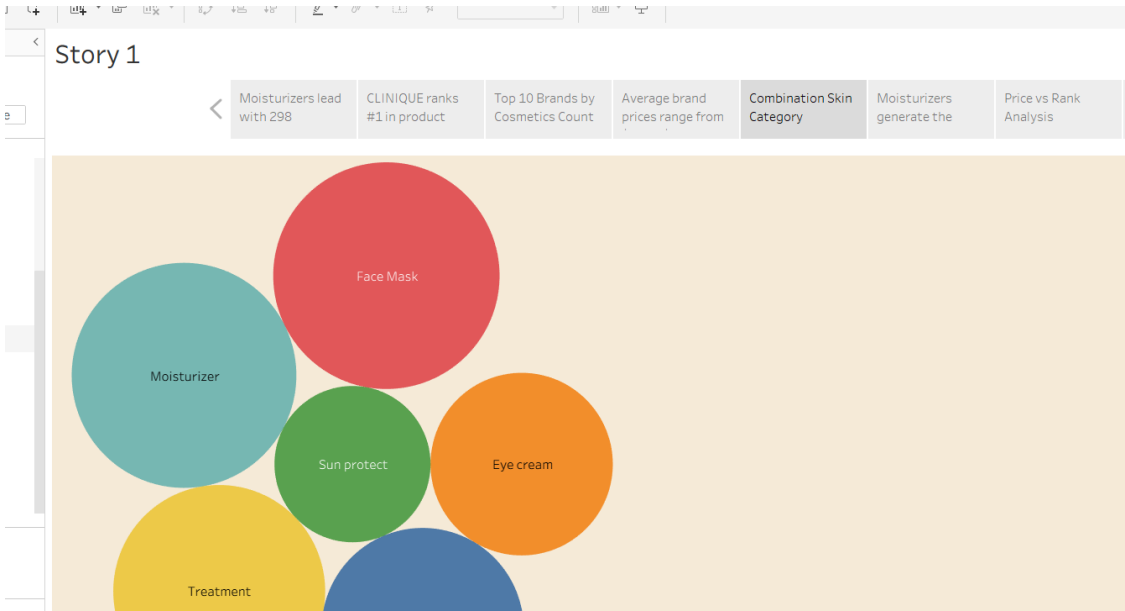


Figure 6: Category-wise Distribution of Combination Skin — Story 1 view

Business Question 5

Which top 10 brands offer the most products suitable for all five skin types simultaneously?

Worksheet: Top 10 Brands that are suitable for all skin types | Chart: Vertical bar chart

Insight	Finding
#1 (all-skin)	DR. JART+ — highest ALL SKIN TYPES total (~150)
Strong performers	KIEHL'S SINCE 1851 (~140), ORIGINS (~120)
Takeaway	DR. JART+ and KIEHL'S are best for universal skin-type coverage

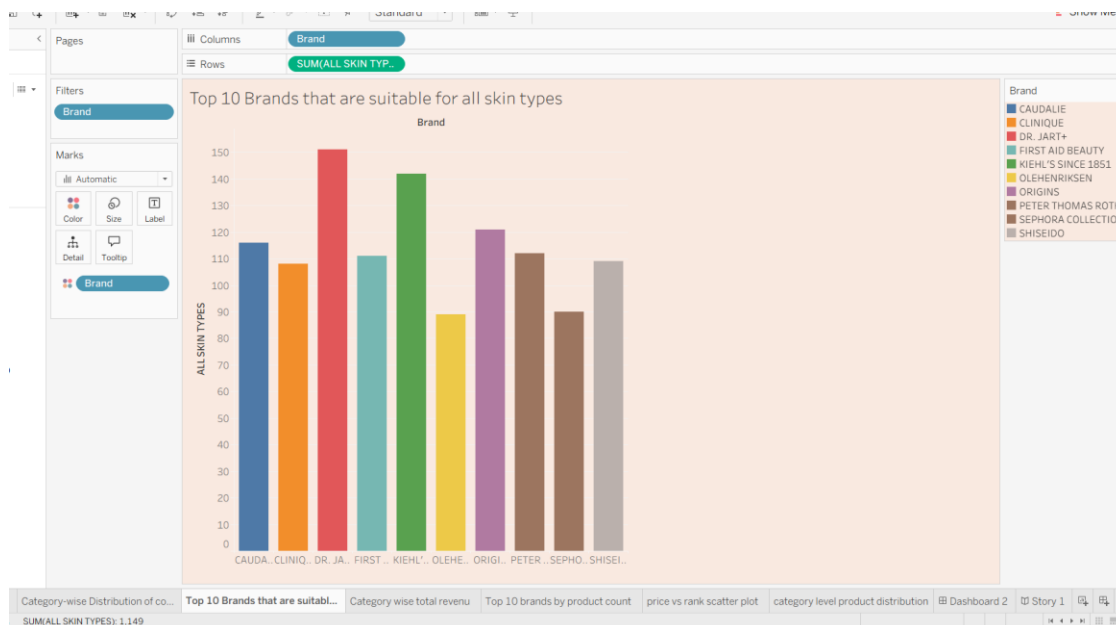


Figure 7: Top 10 Brands suitable for all skin types

Business Question 6

Which product category generates the highest total revenue (price proxy)?

Worksheet: Category wise total revenue | Chart: Horizontal bar chart

Insight	Finding
Highest revenue	Moisturizer — \$20,588 total price
Strong #2	Treatment — \$19,636
Lowest	Sun protect — \$7,804

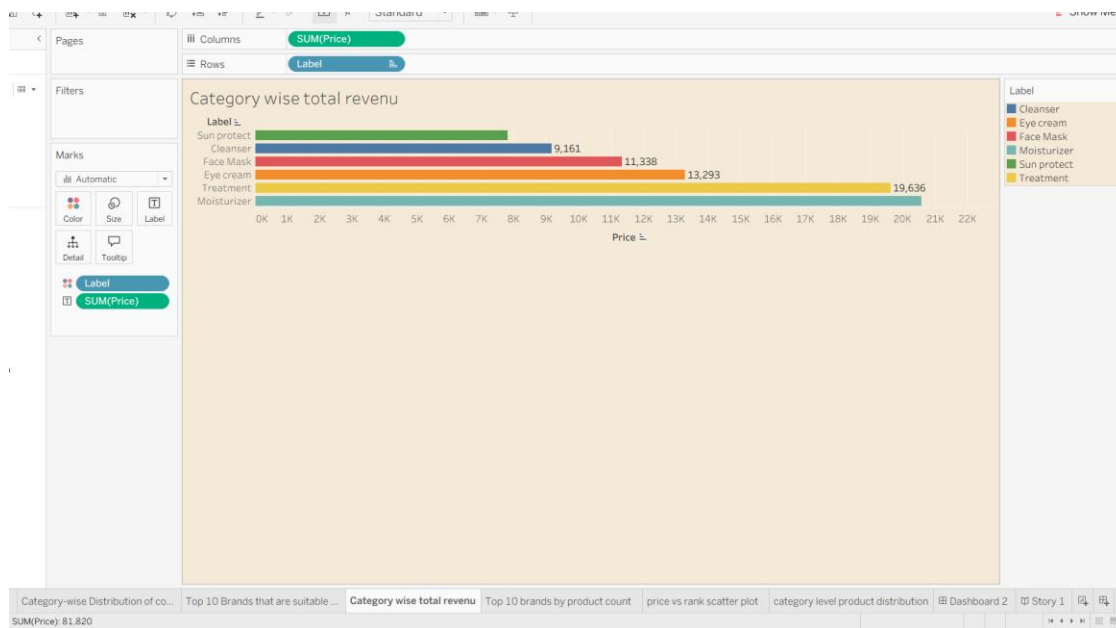


Figure 8: Category wise total revenue

Business Question 7

Which are the top 10 brands by distinct product count?

Worksheet: Top 10 brands by product count | Chart: Horizontal bar chart

Insight	Finding
#1 brand	CLINIQUE — 79 distinct products
Tight mid-pack	SHISEIDO (63), ORIGINS (54), MURAD (47)
#10	KATE SOMERVILLE — 35 products

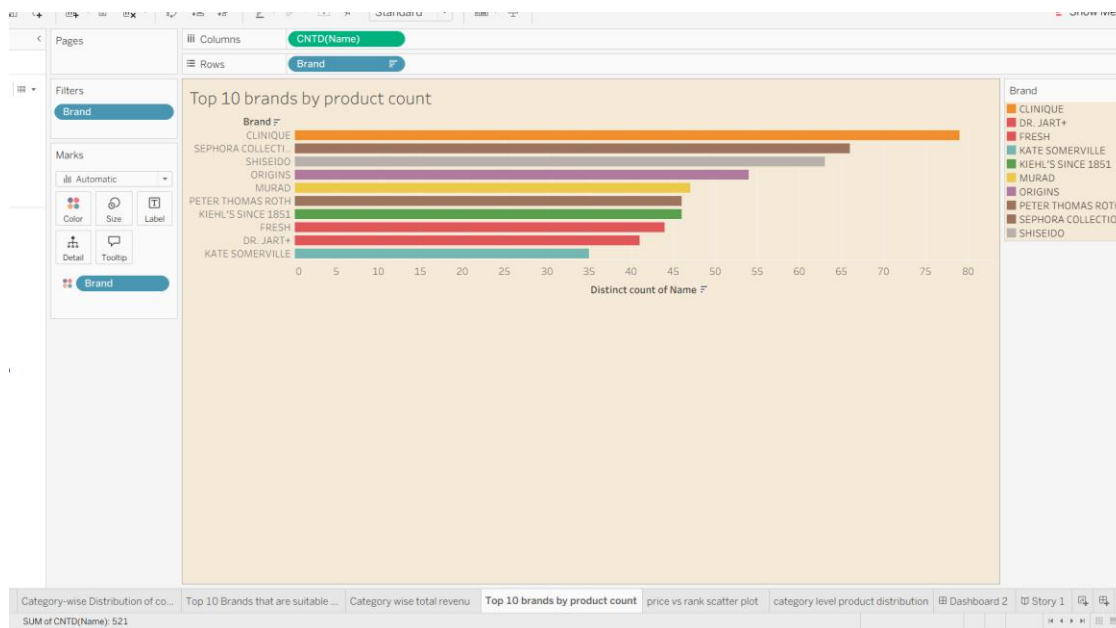


Figure 9: Top 10 brands by product count

Business Question 8

Is there a correlation between price and customer rank — do higher-priced products receive higher ratings?

Worksheet: price vs rank scatter plot | Chart: Scatter plot

Insight	Finding
No strong correlation	High-priced brands do NOT consistently receive higher ratings
Cluster	Most brands cluster in low-price (\$0–\$500) zone
Takeaway	Price is not a reliable predictor of customer satisfaction

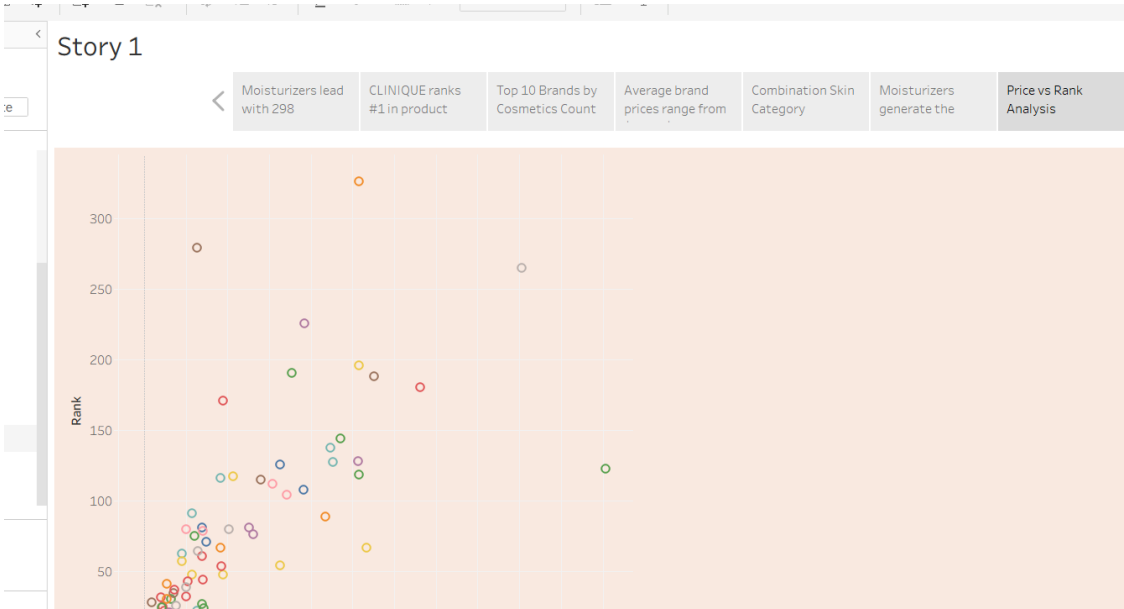


Figure 10: Price vs Rank Analysis — Story 1 view

Business Question 9

How many products does each category contain, and how do they compare?

Worksheet: category level product distribution | Chart: Horizontal bar chart

Insight	Finding
Largest	Moisturizer — 298 products
Runner-up	Cleanser — 281 products
Smallest	Sun protect — 170 products
Total confirmed	1,472 products — matches dataset total exactly

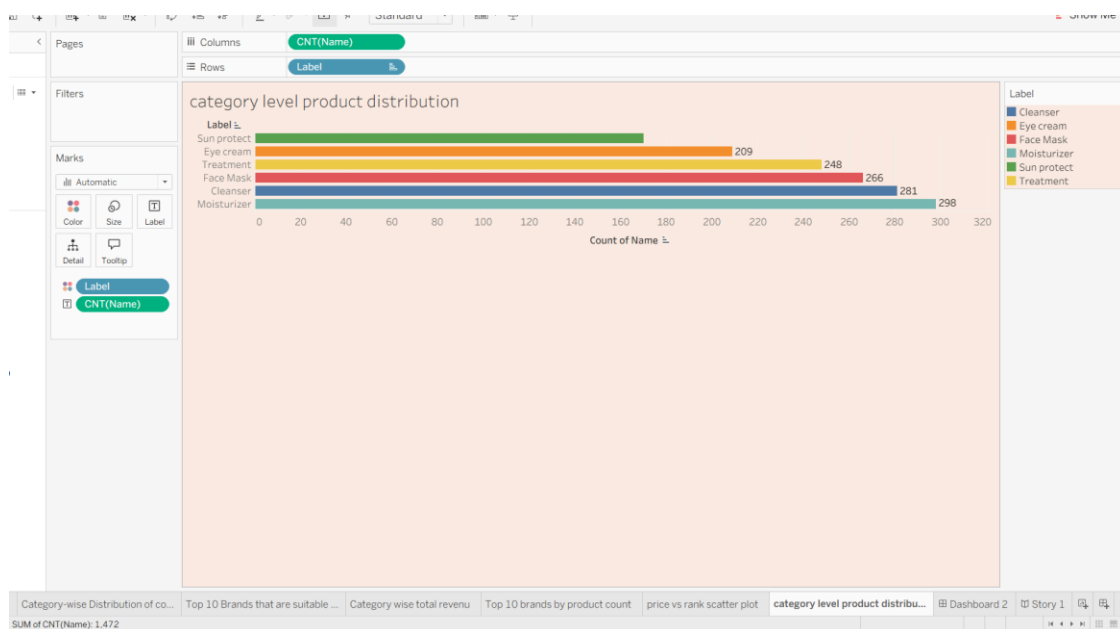


Figure 11: Category level product distribution